

Yifei Dong

Education

- 2025 **Duke University**, *Visiting Scholar*, Robot Dexterity Lab, with Prof. Xianyi Cheng
- 2023 - Now **KTH Royal Institute of Technology**, *PhD Candidate*, Robotics, Computer Science
- 2019 - 2022 **ETH Zurich**, *M.Sc.*, Mechanical Engineering
- 2015 - 2019 **Shanghai Jiao Tong University**, *B.Sc.*, Mechanical Engineering
Tsien Hsue-shen Program & Zhiyuan Honors Program of Engineering

Employment

- 2023 - Now **Robotics, Perception and Learning, KTH, Stockholm**
Doctoral Student, with Assoc. Prof. Florian Pokorny
- 2022 **Idiap, EPFL, Martigny**, *Research Assistant*, with Dr. Sylvain Calinon
Phase estimation on Riemannian manifold. Code contribution: [Robotics Code From Scratch](#)
- 2022 **F&P Robotics, Zürich**, *Software Engineer Intern*
Motion planning of door opening and human-robot interaction for a mobile service robot.
- 2018 **NIO Inc., Shanghai**, *Software Engineer Intern*
LiDAR simulation of an autonomous vehicle in various traffic scenes.

Publications

Journals and Conference Proceedings (* indicates equal contribution):

- [J1] Yifei Dong, Xianyi Cheng, Florian Pokorny, *Characterizing Manipulation Robustness through Energy Margin and Caging Analysis*, **RA-L**, 2024, [Paper](#)
- [J2] Haofei Lu, Yifei Dong, Zehang Weng, Florian Pokorny, Jens Lundell, Danica Kragic, *Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation*, **RA-L**, 2025, [Paper](#) | [Website](#)
- [C1] Kei Ikemura*, Yifei Dong*, Florian Pokorny, *Latent Diffeomorphic Co-Design of End-Effectors for Deformable and Fragile Object Manipulation*, Proceedings of Robotics: Science and Systems (**RSS**), 2026, [Paper](#)
- [C2] Yifei Dong*, Yan Zhang*, Sylvain Calinon, Florian Pokorny, *Robustness-Aware Tool Selection and Manipulation Planning with Learned Energy-Informed Guidance*, **ICRA**, 2026, [Paper](#)
- [C3] Yifei Dong*, Shaohang Han* et al., *CageCoOpt: Enhancing Manipulation Robustness through Caging-Guided Morphology and Policy Co-Optimization*, **IROS**, 2025, [Paper](#)
- [C4] Yifei Dong, Florian Pokorny, *Quasi-static Soft Fixture Analysis of Rigid and Deformable Objects*, **ICRA**, 2024, [Paper](#)
- [C5] David Cáceres Domínguez et al., *The First WARA Robotics Mobile Manipulation Challenge—Lessons Learned*, European Conference on Mobile Robots, 2025, [Paper](#) | [Video](#)
- [C6] Yue Chen et al., *DUET: Joint Exploration of User Item Profiles in Recommendation System*, ACL findings, 2026, [Paper](#)

Under Review, Workshop Paper and Others:

- [1] Yifei Dong, Zhanyi Sun, Lujie Yang, Manuel Baum, Shuran Song, Florian Pokorny, Xianyi Cheng, *Robustness of Robotic Manipulation: Foundations and Frontiers*, under review (IJRR), 2026,
- [2] Kei Ikemura, Yifei Dong et al. *Efficient End-effector Co-Design by Demonstration for Deformable Fragile Object Manipulation*, RSS Workshop on Robot Hardware-Aware Intelligence (Spotlight), 2025, [Paper](#)

- [3] Kei Ikemura*, Yifei Dong* et al., *Sim-to-Real Gentle Manipulation of Deformable and Fragile Objects with Stress-Guided Reinforcement Learning*, tech report, 2025, [Preprint](#)
- [4] Yan Zhang, Yiming Li, Yifei Dong, Florian Pokorný, Sylvain Calinon, *Physics-Informed Eikonal Caging for Whole-Arm Manipulation Planning*, under review (IJRR), 2026,
- [5] David Blanco-Mulero, Yifei Dong, Julia Borrás et al., *T-DOM: A Taxonomy for Robotic Manipulation of Deformable Objects*, under review (RAS), 2025, [Preprint](#) | [Website](#)
- [6] Christoffer K. Øhrstrøm, Rafael I. C. Muchacho, Yifei Dong et al., *Parabolic Position Encoding: Vision-Centric, Principled, Extrapolatable, General*, under review (NIPS), 2026, [Preprint](#)
- [7] Yue Chen et al., *WarriorMath: Empowering Mathematical Reasoning for Large Language Models via Expert Battles*, tech report, 2025, [Preprint](#)

Academic Activities

Talk, Academic Service, Supervision and Teaching:

- 2026 Main organizer of ICRA 2026 Workshop on manipulation robustness, Vienna | [Website](#)
- 2025 Guest lecture at Duke University's graduate course, Robotic Manipulation, 2025 Spring
- 2024 Invited talk at Prof. Joel Burdick's group, California Institute of Technology
- 2024 - Now Reviewer for IJRR, RA-L, ICRA, IROS, ISRR
- 2024 - 2026 Supervised master students, Kei Ikemura (next: KTH), Hongjia Liu (next: UCSD), Li Chen, Gawtam C. Ramesh, Zhenyuan Liang
- 2024, 2025 Teaching assistant in Introduction to Robotics, DD2410, KTH

Experiences

- 2024 **Mobile Manipulation Challenge, ABB, Västerås, Sweden**
Team member of KTH: Custom gripper co-design for multi-task manipulation | [Slides](#)
- 2021 **Robotic Systems Lab, ETH, Zürich, Graduate Researcher**, with Prof. Marco Hutter
Master thesis: Mobile door state estimation and parameter identification | [Thesis](#)
Semester thesis: Contact-implicit MPC for mobile manipulation | [Video](#) | [Thesis](#)
- 2020 **Computer Vision and Geometry Lab & Microsoft, Zürich**
Course project: Object mesh reconstruction using RGB-D cameras. | [Thesis](#) | [Git](#) | [Video](#)
- 2019 **SAIC Motor, Shanghai**
Bachelor thesis: Strategy optimization of autonomous emergency braking
- 2018 **Institute of Design and Control Engineering, SJTU, Shanghai**
3-DOF plum processing system for surface-curving and core-deprivation. | [Video](#)
- 2017 **Institute of Intelligent Vehicle, SJTU, Shanghai**
Structural design and locomotion formulation of a snake robot. | [Video](#)

Awards & Grants

- 2024 Travel Grant from the Karl Engvers Foundation
- 2021, 2022 ETH Scholarship for International Students
- 2019 Excellence Award, BAIC Automobile Invitational Tournament
- 2018 Tan Kah Kee Invention Award
- 2016, 2018 General Motors Scholarship
- 2017 Huawei Scholarship
- 2016, 2017 Zhiyuan College Honors Scholarship

Skills

- Robot Franka Panda, Allegro hand, Ufactory Xarm 7, ANYmal legged robot, etc.
- Programming Python, C++, MATLAB, C, C#
- Tools ROS/ROS2, Genesis, Mujoco, Pybullet, Isaac Gym, Git, Docker, Blender, PyTorch, Codex, Simulink, Unity3D, CAD (ANSYS, SolidWorks, CATIA, NX Unigraphics)
- Language English (TOEFL 109/120), Chinese (native), German (Goethe B1)